



ବଝେଘା RACING

NEWSLETTER - August 2017



LIVE IT. LOVE IT. RACE IT.



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Formula SAE Italy

19th to 23rd July, 2017
Riccardo Paletti Circuit
Varano de' Melegari





Static Events

The team reached the Riccardo Paletti Circuit on Day zero of the event, which was spent in registration and preparing for scrutiny.

The next day, with speedy progress along technical inspection for Scrutiny, Brakes, Tilt and Noise tests, the team cleared these rounds with ease.

The evolution in the design of the car from the previous year helped the team score 90/150 in the Design Event.

The Cost and Business event scores - 69/100 and 42.8/75 respectively, were above average with a considerable scope for improvement.

Dynamic Events

The final two days of the event were dedicated to the dynamic events that test the performance of the car on the tracks.

The team successfully completed the Acceleration, Skidpad, Autocross as well as 22 laps of the endurance event.

The car timed an outstanding 72s lap in Autocross and a 66s lap in the endurance run.

The event saw a great advancement from all the previous results with the team believing in achieving greater heights through small increments of success.

Ashwa Racing bagged an overall position of 25/60 registered teams and hopes to continue and better its performance in the upcoming years with constant unabated support from all its supporters and well wishers.



All's well that ends well



With the end of Formula SAE Italy, yet another race season has reached a successful conclusion. As we look back into the past 12 months of simple joys, magnificent victories, sleepless nights and the occasional miseries, we sigh and grin.

Being a student team reaching for the stars, our path was strewn with rocks and thorns. From lack of funds at crucial times to unforeseen delays in logistics and manufacturing, we've faced everything... (All Hail Edward Murphy!) Indeed, we take pride as a team to have stuck together through thick and thin and to have emerged stronger than ever.

Having had a decent event at Formula Bharat, the combustion team led by Sohan Shetty, Pranjal Tyagi and Gokul Suresh, quickly shifted gears. FS Italy saw Ashwa Racing exhibit its engineering and

managerial prowess with much awe. Learning from previous not-so-kind experiences, the combustion team dashed through 22 laps of Endurance in FS Italy as opposed to the meagre 5 laps we could manage in the Indian event. The Hybrid team was no less, bagging the podium in many events in Formula Hybrid and ultimately becoming the 2nd Overall Champion.

At this bittersweet moment, when we part with our beloved graduating seniors, we thank them for all that they have given us – Guidance, Memories and the most important of all, the Legacy of Ashwa Racing. RZX7, you will be remembered fondly.

As we eagerly look forward to the 2018 race season, we continue to push the boundaries of interdisciplinary engineering. We thank all our parents, the college, the industry partners, the alumni and all our well-wishers for their invaluable support.

Handing over the baton



Sweekruth Shetty
Team Captain
RZ-X8C



Prateek Bhustali
Chief Engineer
RZ-X8C



Rakesh H N
Project Manager
RZ-X8C

The leaders of the RZ-X7H project - Sweekruth, Prateek & Rakesh have led the team successfully in the 2016-17 race season and are gearing up to lead the way for the 2017-18 combustion project, while handing over the leadership responsibility of the hybrid project to three capable team members of the succeeding batch.



Asfan Khan - Team Captain - RZ-X8H

A person who is best described with the word 'relentless'; his level of maturity and passion towards racecars has enabled him grow leaps & bounds in Ashwa Racing. His zeal towards the project and a 'won't stop till the work is done' attitude catches the eye. Asfan has an impressive quality of motivating his teammates and bringing out the best in them, which makes him apt for the role of Team Captain.



Uday Naik - Chief Engineer - RZ-X8H

A dependable and competent member of the team who can be found in the workshop at any time of the day, Uday is a bundle of knowledge, technically equipped to handle intricate aspects of all modules of an FSAE car. The role of a chief engineer at Ashwa Racing, filled with esoteric tasks, has been rightly assigned to him.



Suhas B U - Project Manager - RZ-X8H

Suhas is a keen learner whose positive ways of motivation and ability to keep the mood light, making a difficult task look easy, are truly inspiring. Upon taking up the project management role, he has exhibited a lot of discipline and good conduct with constant reminders for updates and documentation, keeping the whole team on track.

ON THE BASE PLATE



Yet another month has come to an end, with the team spending hours together in the workshop and working meticulously to stay committed to the deadlines of the project.

The 2018 combustion chassis is nearing completion. Welding the clamps on to the chassis, will seal the deal after which it will be taken out of the base plate. This will be followed by the construction of the hybrid chassis for the next race season.

The team is extremely grateful to all its manufacturing partners for extending their support to Ashwa Racing for the 2017-18 race season, ensuring that our quest to make our designs lighter is not constrained by manufacturing challenges.

The required machined components including the hubs provided by DMG MORI have been received, which will be assembled on the car in the upcoming days.

Procurement of the expensive and critical OEM components for the next three year cycle, is in progress, as the team hopes to start the assembly phase of the 2018 prototypes at the earliest.

Welcome Freshers!



As a part of the PR and networking activities, the team set up a display for the freshers on the day of the inauguration providing them with an understanding of the project and the perks of being involved in it. The team also had the opportunity to interact with parents, visitors and Industry professionals, thereby receiving invaluable inputs to the project.



SPONSOR FOCUS



Sansera Engineering is a committed and highly trusted engineering partner of Ashwa Racing offering precision engineered components, and has played a monumental role in the progress of the team over the years. We are extremely grateful to Sansera Engineering for their unending support towards the project on the financial and manufacturing front.

The team is extremely grateful to Deeyem CNC Machining Pvt Ltd for the perfect precision manufacturing of our components. We are in awe of their commitment to our timeline; the quick response of Deeyem has saved us a lot of hassle in our assembly phase. Deeyem has also paid attention to the most minute details, all the way to neat packaging during delivery.



Jindal Aluminium is the leading aluminium extrusion manufacturer in the country and the largest aluminium producer with above 35% of the market share. We thank Jindal Aluminium wholeheartedly for supporting our endeavour and providing raw material for manufacturing of our components.

SPONSOR FOCUS

LIEBHERR

The Liebherr Group is a large equipment manufacturer based in Switzerland. The team would like to thank Liebherr for the manufacturing of toothed gear components for the prototypes. Liebherr has gone a step further and procured high grade aluminium material for the manufacturing. We are extremely grateful to them for their continued support.

Since its foundation in 1997, Magod Laser has consistently built recognized expertise, helping customers to adopt laser based manufacturing processes & applications for numerous industries. We would like to thank Magod Laser for profiling our steel tubes for the manufacture of the chassis and being a constant source of encouragement towards the Ashwa Racing endeavour.



EIGEN ENGINEERING SERVICES

Eigen Engineering Services has manufactured the Motor generator tensioners for the 2018 prototypes. We wholeheartedly thank Prof. M.S. Nagaraj, a visiting faculty at RVCE, for his immense support and wishes to the team.

DMG MORI has provided timely support to us for the upcoming race-season by manufacturing our hubs, which are a set of critical components in the wheel assembly. We thank them for their support and commitment towards the project.

DMG MORI



Manucho Machines Limited has helped the team by manufacturing the Upright Clamps used in the suspension system of the 2018 prototypes.

AUDIT - July 2017



R. V. COLLEGE OF ENGINEERING

अश्वरACING

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R. V. COLLEGE OF ENGINEERING **अश्वर** RACING

Accounts for the period from 1-07-2017 to 29-07-2017

RECIEPTS	AMOUNT
Opening Balance in Kotak Mahindra Bank RVCE Branch	
SB A/c 136010046665	₹ 1,327,616.41
SB A/c 136010058514	₹ 354,290.80
SB A/c 136010060793	₹ 240,739.93
cash in hand	₹ 1,190.00
83H remaining cash in	₹ 6,424.00
italy accomodation	₹ 300,000.00
TOTAL	₹ 2,230,261.14

PAYMENTS BY	AMOUNT
Powertrain & Smp	₹ 111,494.00
Electrical	₹ 80,811.00
Miscellaneous	₹ 345,637.90
Brakes & Logistics	₹ 291,503.00
Chassis & Suspension	₹ 15,891.00
SMS & Wire Charges	₹ 17.70
Cash In Hand	₹ 3,828.00
Closing balance in Kotak Mahindra Bank , RVCE Branch	
SB A/c 136010046665	₹ 1,064,067.71
SB A/c 136010058514	₹ 221,270.90
SB A/c 136010060793	₹ 95,739.93
TOTAL	₹ 2,230,261.14

Utkarsh

Utkarsh Agrawal
Finance Team

अश्वर RACING

R.V. College of engineering

ALUMNUS OF THE MONTH



Rohith, popularly known as GY, joined Ashwa in his freshmen year in December 2012 and was known for his sense of ingenuity, astuteness and a peculiar perspective. He was the Chief Engineer for Ashwa's first Hybrid prototype that raced at the New Hampshire Motorspeedway in 2015.

Ashwa's maiden participation was riddled with challenges which were surmounted with diligence, perseverance and a broad mind. Being awry of dogmatic beliefs was crucial in winning the 7th place and laid a strong foundation for the growth and development of Ashwa Racing's Hybrid venture.

He held a keen interest in CFD (Computational Fluid Dynamics) which drove him to take up the Design and Fabrication of an Aerodynamic package for an FSAE racecar as his final year thesis. [An aerodynamic package is aimed improving the cornering velocity of the car due to increased traction caused by the downforce generated by the wing elements]. The thesis brought fairly new ideas to the table about Aerodynamic Packages. The thesis will help the future batches breakdown the cumbersome task of developing aerodynamic packages for the racecar.

Rohith graduated in July 2016 and is currently employed by Seagate as a Structural/Thermal Design Engineer.

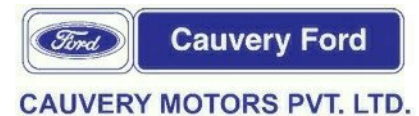
On a lighter note, under the charade of a fancy title in an MNC like Seagate, Rohith is basically responsible for fanning hard drives. Also, everyone's been expecting him to supply hard drives, but he has been covertly turning them down.



2016-17 COMBUSTION PARTNERS



2016-17 SERVICE PARTNERS



2016-17 COMBUSTION - PRODUCT PARTNERS

**BHUMIPUTRA
ARCHITECTURE**

Continental 



**MOUSER
ELECTRONICS**



FAG

SCHAEFFLER GRUPPE

MOTUL

molex®



 **Suprajit**

 **bhatia aerospace**
bhatia tools & technology

UF UNIFLEX

 **TE**
connectivity

 **Jupiter**®

FESTO



LOCTITE

BAKER
....a passion for precision



**MANN+
HUMMEL**



M MIRANDA TOOLS

2016-17 COMBUSTION - MANUFACTURING PARTNERS



DMG MORI

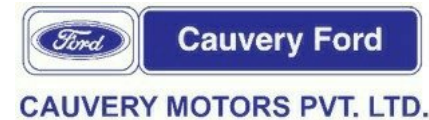


MALLAR
GROUP



www.icp-india.com

SANSERA
ideas@work



**EIGEN ENGINEERING
SERVICES**

PRAGATI
Partnering Machine Tool Builders Worldwide

LIEBHERR



**Srinu Engineering
Industries**



HI-FLEX TECHNOLOGIES

AGNI MOTORS



MECKON



auma
Solutions for a world in motion

VENKATESHWARA MANUFACTURING SYSTEMS

SUN PRECISION TOOLS



2016-17 HYBRID PARTNERS

Life Is On

Schneider
Electric

AMS[®]
In Pursuit of Excellence

BFW

infineon

PEGASYS

SOLVAY
asking more from chemistry[®]

ANW

سابک
sebic

DURAG

**GRIND
MASTER**

DANA

SPICER[®]
Drivetrain Products

Phillips
MACHINE TOOLS INDIA

Werner Finley[®]
Process Cooling Since 1986

DuFlon[™]
Engineering Excellence

BILL FORGE
YOUR PROACTIVE PARTNER

RÖHM
driven by technology

2016-17 SERVICE PARTNERS

Wrike

RICARDO

Self Ride Bikes
RB

ANSYS[®]

joe's
CUSTOMERS | PERFORMANCE TUNING | SERVICE

SCV
SOLUTIONS INC

MERITOR

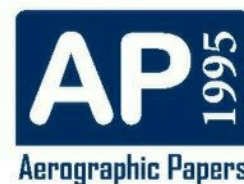
RACE dynamics
Performance Through Innovation

**MANN +
HUMMEL**

2016-17 HYBRID- PRODUCT PARTNERS



BUILDING TRUST



2016-17 HYBRID - MANUFACTURING PARTNERS



DMG MORI



MALLAR
GROUP



FIe TECH
Fantasy in Fibreglass



Cauvery Ford

CAUVERY MOTORS PVT. LTD.



**EIGEN ENGINEERING
SERVICES**



LIEBHERR



**BHAGAVATHI
TOOLINGS PVT. LTD.**



SHRI VENKATESHWARA HI-TECH

**Srinu Engineering
Industries**



MANUCHO MACHINES (P) LIMITED
ISO 9001-2008 COMPANY

HI-FLEX TECHNOLOGIES



AGNI MOTORS



auma
Solutions for a world in motion



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Chassis : Brian N Quadras	Workshop : Arunraj Subbaraj
Suspension : Kanal Sebastian	Admin : Arunraj Subbaraj
Brakes : Vamsi Naramsetty	Logistics : Pallav Rawat
Steering : Eshwarmurthy T	Finance : Mohammad Hannan
Engine : R Manjeet	SMP : Karan Chatrath
Drivetrain : Rajat Kabra	HR : Tejas Babaria
Electrical : Siva Subramanian	Testing : Rajit Malik
Coding : Arunraj Subbaraj	IT : Arunraj Subbaraj

STATIC EVENT MODULE COORDINATOR

Design : Arunraj Subbaraj	PMP : Arunraj Subbaraj
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